

DATA ANALYSIS REPORT

Project / Case Study: Cyclistic Ride Share: From Casual to Annual

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Date: 09/22/2026

EXECUTIVE SUMMARY

This case study analyzes Cyclistic bike-share data from September 2025 through August 2026 to understand how casual riders and annual members use the service differently. The goal is to identify patterns that can help Cyclistic develop strategies to convert more casual riders into annual members.

The analysis found that annual members accounted for about 65% of all rides, while casual riders accounted for about 35%. Members rode more consistently during the workweek and showed strong peaks around typical commuting hours, especially in the morning and late afternoon. In contrast, casual riders were more active on weekends and generally took longer trips, averaging 20.70 minutes compared with 12.37 minutes for members.

Based on these findings, Cyclistic should target casual riders with membership promotions during weekends, emphasize the value of membership to riders who take longer trips, and promote annual membership as a convenient option for regular weekday transportation. These strategies focus on the main behavioral differences identified in the analysis and could help Cyclistic encourage more casual riders to become annual members.

1. BUSINESS TASK

Business Problem

Cyclistic wants to increase the number of annual memberships because annual members are more profitable than casual riders. The marketing team believes there is an opportunity to convert existing casual riders into annual members but first needs to better understand how the two groups use Cyclistic bikes differently.

Business Objective

The objective of this analysis is to examine Cyclistic's historical bike-trip data and identify differences in riding behavior between casual riders and annual members. These insights will help the marketing team develop data-driven strategies aimed at converting casual riders into annual members.

Key Question(s)

- How do annual members and casual riders use Cyclistic bikes differently?
- Why would casual riders buy Cyclistic annual memberships?
- How can Cyclistic use digital media to influence casual riders to become members?

Stakeholders

The primary stakeholders are **Lily Moreno, Cyclistic's Director of Marketing; the Cyclistic marketing analytics team; and the Cyclistic executive team.** The marketing team will use

the findings to develop its strategy, while the executive team will ultimately decide whether to approve the recommended marketing program.

2. DATA SOURCES

Data Source 1 — 202509-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: September 2025

Size: 714,759 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for September 2025. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 2 — 202510-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: October 2025

Size: 646,039 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for October 2025. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 3 — 202511-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: November 2025

Size: 356,628 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for November 2025. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 4 — 202512-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: December 2025

Size: 140,534 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for December 2025. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 5 — 202601-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: January 2026

Size: 137,787 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for January 2026. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 6 — 202602-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: February 2026

Size: 201,450 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for February 2026. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 7 — 202603-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: March 2026

Size: 317,037 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for March 2026. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 8 — 202604-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: April 2026

Size: 448,252 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for April 2026. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 9 — 202605-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: May 2026

Size: 653,704 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for May 2026. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 10 — 202606-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: June 2026

Size: 762,550 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for June 2026. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 11 — 202607-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: July 2026

Size: 869,051 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for July 2026. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Source 12 — 202608-divvy-tripdata

Source: Divvy Historical Trip Data, City of Chicago

Format: CSV

Time Period: August 2026

Size: 868,191 rows and 13 columns

Description: Contains individual Divvy bike-share trip records for August 2026. The dataset includes ride IDs, bike type, trip start and end times, start and end stations, geographic coordinates, and rider type (member or casual). This information is useful for comparing riding patterns between casual riders and annual members.

Data Reliability and Limitations

The Cyclistic trip data is reasonably credible because it comes from Divvy's historical trip records and is made publicly available for analysis. The case study states that the data is provided by Motivate International Inc. under its data license. It also notes that personally identifiable rider information is not available because of privacy restrictions. As a result, the analysis cannot determine characteristics such as where riders live or whether the same casual rider purchased multiple single-use passes.

The 12 monthly datasets use a consistent 13-column structure, and important fields such as *ride_id*, *rideable_type*, *started_at*, *ended_at*, *start_lat*, *start_lng*, and *member_casual* contain no missing values. However, missing data is more common in the station fields. Across the monthly files, approximately **16%–22%** of *start_station_name* and *start_station_id* values are missing, while about **17%–24%** of *end_station_name* and *end_station_id* values are missing. Missing values for *end_lat* and *end_lng* are much smaller, generally below **0.15%**.

Another limitation is that the dataset records trip behavior, but it does not explain **why** riders behave differently. It also lacks demographic information, membership purchase history, or other personal characteristics that could help explain why casual riders might choose to become annual members. Therefore, the analysis can identify patterns in bike usage, but it cannot directly determine rider motivations.

Because the data represents recorded Cyclistic/Divvy trips rather than a survey sample, traditional sampling bias is less of a concern. However, the analysis is still limited to trips that were successfully recorded by the system and to the information included in the dataset.

Missing station information and privacy-related restrictions should therefore be considered when interpreting the results.

3. DATA CLEANING AND MANIPULATION

Tools Used

PostgreSQL, accessed through DBeaver, was used as the primary tool for combining, inspecting, cleaning, and transforming the 12 monthly trip datasets. PostgreSQL was appropriate because the combined dataset contained more than six million records and was too large to work with efficiently in spreadsheet software. Python was used to independently validate the cleaned dataset and will also support later analysis. Tableau will be used to create visualizations during the Share phase.

Cleaning Process

- **Checked for duplicate records:** No duplicate *ride_id* values were found within the individual monthly datasets. After the 12 datasets were combined, 35 duplicated *ride_id* values were identified. Each appeared twice and contained identical trip information. The duplicates occurred across the April 2026 and May 2026 datasets, so one copy of each duplicated trip was retained in the cleaned dataset.
- **Handled missing or null values:** The critical fields *ride_id*, *started_at*, *ended_at*, *rideable_type*, and *member_casual* contained no missing values. Missing values were present in some station-related fields, but these records were retained because they could still be used for analyses that did not require station information.
- **Checked data types and formatting:** Date and time fields were reviewed to ensure they could be used for duration and date-based calculations. Categorical fields were also checked for inconsistent spelling, unexpected values, or blank categories. No inconsistencies were found in *member_casual* or *rideable_type*.
- **Investigated invalid values and outliers:** Twenty-nine records in the November 2025 dataset produced negative ride durations because *ended_at* occurred before *started_at*. The records themselves were retained because other trip information remained usable, but their calculated *ride_length* values were set to NULL so that the invalid durations would not affect duration-based analysis. Unusually long rides were also investigated rather than automatically removed.
- **Created calculated fields:** A *ride_length* variable was created by subtracting *started_at* from *ended_at*. A *day_of_week* variable was created from *started_at*, using 1 for Sunday through 7 for Saturday as specified in the case study.
- **Combined the datasets:** The 12 monthly datasets were combined using UNION ALL into a full-year raw table. A *source_month* column was added to identify the original monthly dataset for each trip. The original monthly tables and the combined raw table were preserved so that the source data remained unchanged.

Quality Checks

The combined raw dataset contained **6,115,982 rows**. After removing 35 duplicate copies, the cleaned dataset contained **6,115,947 unique trips**. SQL validation confirmed that no duplicate *ride_id* values or negative *ride_length* values remained. The 29 records with invalid negative durations were successfully represented as NULL for *ride_length*, and *day_of_week* contained only valid values from 1 through 7.

Python was then connected directly to the PostgreSQL database to independently verify the cleaned dataset. The Python validation confirmed **6,115,947 total rows, 6,115,947 unique ride**

IDs, 29 missing ride-length values, zero negative ride lengths, and day-of-week values ranging from 1 to 7. Based on these checks, the dataset was considered sufficiently cleaned and structured for the Analyze phase.

4. ANALYSIS

Approach

PostgreSQL and Python were used to analyze differences between casual riders and annual members during the September 2025 through August 2026 period. Descriptive statistics, grouped ride counts, percentages, averages, medians, and measures of variation were used to compare rider behavior. The analysis focused on ride duration, day of week, month, time of day, bike type, and starting-station patterns because these variables directly help answer how casual riders and members use Cyclistic differently. Python was also used to validate the SQL results and perform additional comparisons.

Analysis Steps

1. **Compared overall riding behavior by rider type.** Ride counts, average ride duration, and median ride duration were calculated for casual riders and members. Both the mean and median were examined because unusually long rides could increase the average.
2. **Analyzed when each rider type uses Cyclistic.** Ride activity was compared by day of week, month, hour of day, and broader time-of-day categories. Percentages within each rider type were used so that casual and member patterns could be compared fairly despite different total ride counts. Seasonal variation was also measured using peak-to-low ratios and coefficients of variation.
3. **Compared bike and location usage.** Bike-type usage was analyzed as a percentage of rides for each rider group. The most frequently used starting stations were also identified, and the share of known-station rides represented by the top 10 stations was calculated. SQL results were then checked and summarized in Python.

Results

The analysis showed several clear differences between casual riders and annual members. Casual riders generally took longer trips, averaging **20.66 minutes** with a median of **10.86 minutes**, compared with an average of **12.37 minutes** and a median of **8.55 minutes** for members. Casual riding was also more concentrated on weekends: **37.26% of casual rides occurred on Saturday or Sunday**, compared with **23.34% of member rides**. Members were more active during weekdays, particularly in the middle of the week.

Casual ridership also showed greater seasonal variation. The busiest casual month had **14.46 times** as many rides as the least busy month, compared with a ratio of **4.63** for members. The coefficient of variation was also higher for casual riders (**0.67**) than members (**0.44**), showing that casual ridership changed more sharply throughout the year.

Time-of-day patterns were also different. Both rider groups reached their highest hourly ride count around **5:00 PM**, but members had substantially more morning activity. **25.18% of member rides occurred during the morning**, compared with **15.56% of casual rides**, while casual riders were more concentrated during midday and night.

Electric bikes accounted for most rides for both groups, but casual riders used them for a somewhat larger share of trips: **73.60% compared with 68.06% for members**. Starting-station

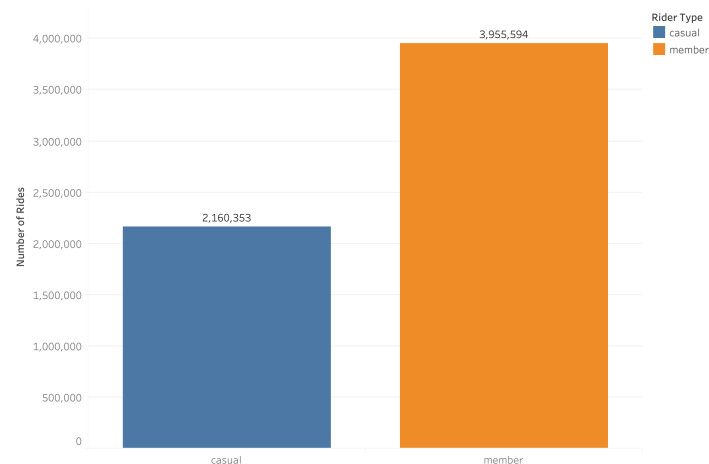
patterns also differed. The top 10 casual starting stations accounted for **12.17% of casual rides with a known starting station**, compared with **6.12% for members**, suggesting that casual rides were more concentrated around a smaller number of high-use locations.

Overall, the results indicate that casual riders tend to take longer rides, ride more heavily on weekends, show greater seasonal variation, and concentrate more around certain starting locations. Members use Cyclistic more consistently throughout the week and year and show stronger morning usage. These patterns provide the foundation for the visualizations and recommendations developed in the next phases of the case study.

5. SUPPORTING VISUALIZATIONS AND KEY FINDINGS

Figure 1. Total Rides by Rider Types

Members account for nearly two-thirds of Cyclistic rides



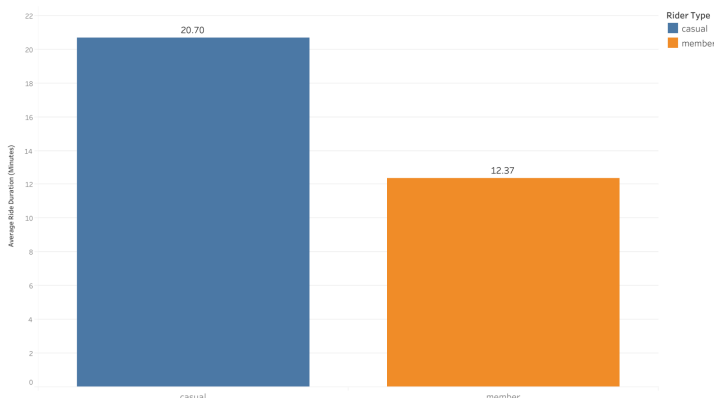
Key Finding

Members account for nearly two-thirds of all Cyclistic rides.

Members completed **3,955,594 rides (64.68%)**, while casual riders completed **2,160,353 rides (35.32%)**. Out of more than 6.1 million rides analyzed, members made substantially more trips overall. This suggests that members use the bike-share service more frequently than casual riders.

Figure 2. Average Ride Duration by Rider Type

Casual riders take longer trips than members



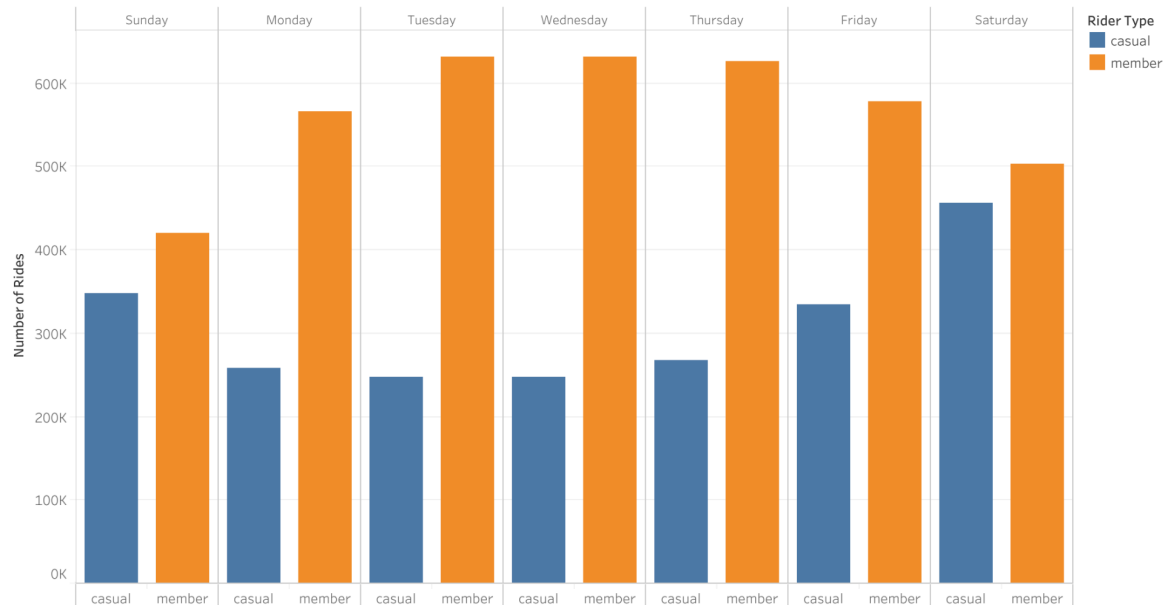
Key Finding

Casual riders take longer trips than members.

Casual riders averaged **20.70 minutes per ride**, compared with **12.37 minutes for members**, a difference of about **8.33 minutes**. The median ride duration was also longer for casual riders at **10.86 minutes**, compared with **8.55 minutes for members**. This shows that although members take more rides overall, casual riders tend to spend more time on each individual trip.

Figure 3. Rides by Day of Week

Casual riders concentrate more rides on weekends, while members ride most midweek



Key Finding

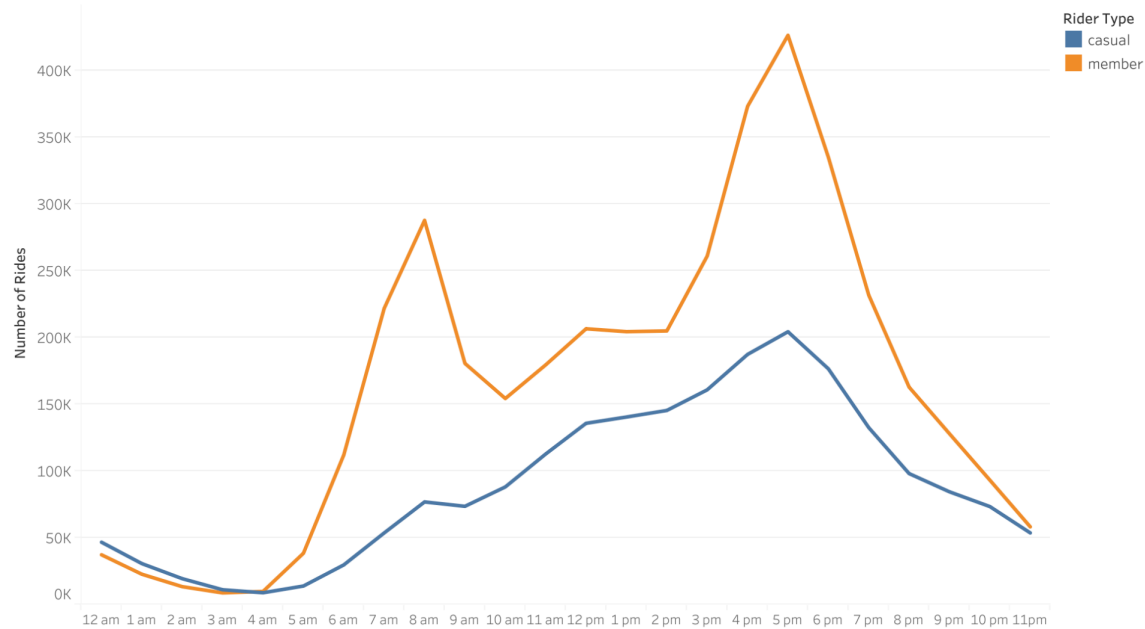
Casual riding is more concentrated on weekends, while member riding is strongest during the middle of the workweek.

Saturday was the most popular day for casual riders, accounting for **456,663 rides (21.14%)** of all casual trips. Sunday accounted for another **16.13%**, meaning approximately **37.27% of casual rides occurred on the weekend**.

Member usage followed a different pattern. **Wednesday had the most member rides at 631,953**, followed closely by Tuesday at 630,934 and Thursday at 625,494. Only about **23.34% of member rides occurred on Saturday and Sunday combined**. This suggests that member usage is more strongly associated with regular weekday travel, while casual riding is more concentrated around weekends.

Figure 4. Rides by Hour of Day

Member rides peak around 8 AM and 5 PM, while casual rides are more evenly distributed



Key Finding

Members show strong morning and evening usage peaks, while casual riding increases more gradually throughout the day.

Member rides rise sharply during the morning, reaching **287,997 rides at 8 AM**, before declining and then reaching their largest hourly total of **426,718 rides at 5 PM**. These two distinct peaks are consistent with common morning and evening travel periods.

Casual riders show a less pronounced morning peak. Their usage generally increases throughout the day and reaches its highest point at **5 PM with 204,507 rides**. The difference suggests that members may use Cyclistic more regularly for scheduled transportation, while casual riders have a more flexible daily riding pattern.

Figure 5. Monthly Ride Trends

Members ride more year-round, while both groups follow a strong seasonal pattern



Key Finding

Both groups show strong seasonal riding patterns, while members record more rides in every month analyzed.

Ride activity decreased considerably during the winter and increased again through the spring and summer. Casual rides fell from **265,268 in September 2025** to a low of **24,740 in January 2026**, before reaching a high of **357,779 in July 2026**.

Member rides followed the same general seasonal pattern, decreasing to **112,455 in December 2025** before increasing to a high of **521,066 in August 2026**. Members recorded more rides than casual riders in every month of the 12-month period. This indicates that seasonality strongly affects both groups, but member usage remains consistently higher throughout the year.

Overall Key Findings

- **Members are the more frequent users**, accounting for **64.68% of all rides**, compared with 35.32% for casual riders.
- **Casual riders take longer trips**, averaging **20.70 minutes** compared with 12.37 minutes for members.
- **Casual riders are more weekend-oriented**, with about **37.27% of their rides occurring on Saturday or Sunday**, compared with about 23.34% for members.
- **Member usage shows strong 8 AM and 5 PM peaks**, which is consistent with more routine or scheduled transportation patterns.
- **Both rider groups are highly seasonal**, with usage falling during winter and increasing substantially during spring and summer; members nevertheless recorded more rides in every month analyzed.

6. TOP THREE RECOMMENDATIONS

Recommendation 1 — Target Casual Riders on Weekends

Recommendation:

Cyclistic should concentrate membership-conversion promotions on weekends, especially Saturdays and Sundays, when casual riders are most active.

Evidence:

The analysis showed that casual riders have a much stronger weekend riding pattern than annual members. Approximately **37% of casual rides occurred on weekends**, compared with about **23% of member rides**. Saturday was especially important, accounting for about **21% of all casual rides**.

Expected Benefit:

Weekend-focused promotions could allow Cyclistic to reach casual riders while they are actively using the service. Membership offers through the Cyclistic app, email, social media, or other digital channels could increase awareness of annual membership among riders who already use Cyclistic regularly on weekends.

Recommendation 2 — Target Casual Riders Who Take Longer Trips

Recommendation:

Cyclistic should develop membership promotions aimed at casual riders who take longer rides and emphasize the advantages of using Cyclistic more frequently.

Evidence:

Casual riders took substantially longer trips than annual members. Casual rides averaged approximately **20.7 minutes**, compared with about **12.4 minutes for members**. Casual riders also had particularly long average rides on weekends, reaching approximately **24.3 minutes on Sunday** and **23.2 minutes on Saturday**.

Expected Benefit:

Casual riders taking longer trips are already making substantial use of the bike-share service. Targeting these riders with membership information could encourage them to consider whether annual membership would better fit their continued or more frequent use of Cyclistic.

Recommendation 3 — Promote Membership for Regular Weekday Transportation

Recommendation:

Cyclistic should test campaigns that present annual membership as a convenient option for regular weekday transportation.

Evidence:

Annual members displayed a much stronger weekday riding pattern than casual riders. Member activity also showed noticeable peaks around **8:00 AM and 5:00 PM**, while casual riding was more evenly distributed throughout the day. This suggests that annual members use Cyclistic more consistently during periods associated with regular daily transportation.

Expected Benefit:

Showing casual riders how Cyclistic can be incorporated into regular weekday travel could encourage them to use the service more frequently. Increased routine use may make annual membership more relevant to riders who currently use Cyclistic only occasionally or primarily for leisure.

CONCLUSION

The purpose of this analysis was to determine how annual members and casual riders use Cyclistic bikes differently so that the company can develop strategies to encourage more casual riders to become annual members.

The analysis identified several important differences between the two groups. Annual members accounted for approximately **65% of total rides**, while casual riders represented about **35%**. Members rode more consistently during weekdays and showed strong activity around morning and evening travel periods. Casual riders were more concentrated on weekends and took considerably longer rides on average.

Based on these findings, Cyclistic should focus membership-conversion efforts on **weekend casual riders, riders taking longer trips, and casual riders who could potentially use Cyclistic for regular weekday transportation**. These recommendations use the observed differences in riding behavior to identify situations in which membership marketing may be most relevant to casual riders.

The analysis does **not directly determine why an individual casual rider would purchase a membership or which specific digital advertising strategy would be most effective**. Those were broader questions in the original marketing program, while your assigned analysis focused specifically on differences in rider behavior.

APPENDIX

Appendix A — Prepare Phase: Data Inspection

The following SQL queries were used to inspect the 12 monthly Divvy datasets, verify their structure, check row counts, identify missing values, and evaluate data quality before cleaning.

A.1 Verify table structure and data types

```
SELECT
table_name,
ordinal_position,
column_name,
data_type
FROM information_schema.columns
WHERE table_schema = 'public'
    and table_name like 'divvy_trips_%'
ORDER BY table_name, ordinal_position;
```

A.2 — Verify Number of Columns

```
SELECT
```

```

    table_name,
    COUNT(*) AS column_count
FROM information_schema.columns
WHERE table_schema = 'public'
    AND table_name LIKE 'divvy_trips_%'
GROUP BY table_name
ORDER BY table_name;

```

A.3 — Verify Column Names and Data Types

```

select
    ordinal_position,
    column_name,
    data_type,
    count(*) as tables_with_column
from information_schema."columns"
where table_schema = 'public'
    and table_name like 'divvy_trips_%'
group by
    ordinal_position,
    column_name,
    data_type
order by ordinal_position;

```

A.4 — Check Row Counts

```

SELECT '2025_09' AS month, COUNT(*) AS row_count FROM divvy_trips_2025_09
UNION ALL
SELECT '2025_10', COUNT(*) FROM divvy_trips_2025_10
UNION ALL
SELECT '2025_11', COUNT(*) FROM divvy_trips_2025_11
UNION ALL
SELECT '2025_12', COUNT(*) FROM divvy_trips_2025_12
UNION ALL
SELECT '2026_01', COUNT(*) FROM divvy_trips_2026_01
UNION ALL
SELECT '2026_02', COUNT(*) FROM divvy_trips_2026_02
UNION ALL
SELECT '2026_03', COUNT(*) FROM divvy_trips_2026_03
UNION ALL
SELECT '2026_04', COUNT(*) FROM divvy_trips_2026_04
UNION ALL
SELECT '2026_05', COUNT(*) FROM divvy_trips_2026_05
UNION ALL
SELECT '2026_06', COUNT(*) FROM divvy_trips_2026_06
UNION ALL
SELECT '2026_07', COUNT(*) FROM divvy_trips_2026_07
UNION ALL
SELECT '2026_08', COUNT(*) FROM divvy_trips_2026_08

```

```
ORDER BY month;
```

A.5 — Check Date Range of Monthly Data

```
SELECT
  MIN(started_at) AS earliest Ride,
  MAX(started_at) AS latest Ride
FROM divvy_trips_2025_12;
SELECT COUNT(*) AS august_rides_in_september_file
FROM divvy_trips_2025_09
WHERE started_at < '2025-09-01';
```

A.6 — Count Rides Outside the Expected Month

```
SELECT COUNT(*) AS august_rides_in_september_file
FROM divvy_trips_2025_09
WHERE started_at < '2025-09-01';
```

A.7 — Check Missing Values

```
SELECT
  COUNT(*) AS total_rows,
  COUNT(*) - COUNT(ride_id) AS null_ride_id,
  COUNT(*) - COUNT(rideable_type) AS null_rideable_type,
  COUNT(*) - COUNT(started_at) AS null_started_at,
  COUNT(*) - COUNT(ended_at) AS null_ended_at,
  COUNT(*) - COUNT(start_station_name) AS null_start_station_name,
  COUNT(*) - COUNT(start_station_id) AS null_start_station_id,
  COUNT(*) - COUNT(end_station_name) AS null_end_station_name,
  COUNT(*) - COUNT(end_station_id) AS null_end_station_id,
  COUNT(*) - COUNT(start_lat) AS null_start_lat,
  COUNT(*) - COUNT(start_lng) AS null_start_lng,
  COUNT(*) - COUNT(end_lat) AS null_end_lat,
  COUNT(*) - COUNT(end_lng) AS null_end_lng,
  COUNT(*) - COUNT(member_casual) AS null_member_casual
FROM divvy_trips_2025_09;
```

A.8 — Calculate Percentage of Missing Values

```
SELECT
  COUNT(*) AS total_rows,
  ROUND(
    100.0 * (COUNT(*) - COUNT(start_station_name)) / COUNT(*),
    2
  ) AS pct_null_start_station_name,
  ROUND(
    100.0 * (COUNT(*) - COUNT(start_station_id)) / COUNT(*),
    2
  ) AS pct_null_start_station_id,
```

```

ROUND(
  100.0 * (COUNT(*) - COUNT(end_station_name)) / COUNT(*),
  2
) AS pct_null_end_station_name,
ROUND(
  100.0 * (COUNT(*) - COUNT(end_station_id)) / COUNT(*),
  2
) AS pct_null_end_station_id,
ROUND(
  100.0 * (COUNT(*) - COUNT(end_lat)) / COUNT(*),
  2
) AS pct_null_end_lat,
ROUND(
  100.0 * (COUNT(*) - COUNT(end_lng)) / COUNT(*),
  2
) AS pct_null_end_lng
FROM divvy_trips_2025_09;

```

Appendix B — Process Phase: Data Cleaning and Manipulation

B.1 — Check for Duplicate Ride IDs

```

select
  ride_id,
  count(*) as occurrences
from divvy_trips_2025_09
group by ride_id
having count(*) > 1
order by occurrences desc;

```

B.2 — Check Critical Columns for Missing Values

```

SELECT
  COUNT(*) AS total_rows, --counting the total rows in the table--
  COUNT(*) FILTER (WHERE ride_id IS NULL) AS missing_ride_id,
  COUNT(*) FILTER (WHERE started_at IS NULL) AS missing_started_at,
  COUNT(*) FILTER (WHERE ended_at IS NULL) AS missing_ended_at,
  COUNT(*) FILTER (WHERE rideable_type IS NULL) AS missing_rideable_type,
  COUNT(*) FILTER (WHERE member_casual IS NULL) AS missing_member_casual
FROM divvy_trips_2026_08;

```

B.3 — Check for Invalid Ride Durations

```

select
  COUNT(*) AS invalid_time_records
from divvy_trips_2026_08
where ended_at <= started_at;

```

```
SELECT
  COUNT(*) FILTER (WHERE ended_at < started_at) AS negative_duration,
  COUNT(*) FILTER (WHERE ended_at = started_at) AS zero_duration
FROM divvy_trips_2025_11;
```

B.4 — Inspect Ride Duration Range

```
select
  MIN(ended_at - started_at) as minimum_ride,
  MAX(ended_at - started_at) as maximum_ride,
  AVG(ended_at - started_at) as average_ride
from divvy_trips_2025_11;
```

B.5 — Check Categorical Values

```
SELECT
  member_casual,
  COUNT(*)
FROM divvy_trips_2025_09
GROUP BY member_casual
ORDER BY COUNT(*) DESC;
```

B.6 — Combine the 12 Monthly Tables

```
CREATE TABLE divvy_trips_combined_raw AS
SELECT '2025-09' AS source_month, *
FROM divvy_trips_2025_09
UNION ALL
SELECT '2025-10', *
FROM divvy_trips_2025_10
UNION ALL
SELECT '2025-11', *
FROM divvy_trips_2025_11
UNION ALL
SELECT '2025-12', *
FROM divvy_trips_2025_12
UNION ALL
SELECT '2026-01', *
FROM divvy_trips_2026_01
UNION ALL
SELECT '2026-02', *
FROM divvy_trips_2026_02
UNION ALL
SELECT '2026-03', *
FROM divvy_trips_2026_03
UNION ALL
SELECT '2026-04', *
FROM divvy_trips_2026_04
UNION ALL
```

```

SELECT '2026-05', *
FROM divvy_trips_2026_05
UNION ALL
SELECT '2026-06', *
FROM divvy_trips_2026_06
UNION ALL
SELECT '2026-07', *
FROM divvy_trips_2026_07
UNION ALL
SELECT '2026-08', *
FROM divvy_trips_2026_08;

```

```

SELECT
    source_month,
    COUNT(*) AS number_of_rows
FROM divvy_trips_combined_raw
GROUP BY source_month
ORDER BY source_month;

```

B.7 — Identify Duplicate Ride IDs After Combining Tables

```

SELECT
    ride_id,
    COUNT(*) AS occurrences
FROM divvy_trips_combined_raw
GROUP BY ride_id
HAVING COUNT(*) > 1
ORDER BY occurrences DESC;

```

B.8 — Create the Cleaned Dataset

```

CREATE TABLE divvy_trips_clean AS
WITH deduplicated AS (--keeps one row per ride_id--
    SELECT DISTINCT ON (ride_id)
        *
    FROM divvy_trips_combined_raw
    ORDER BY ride_id, source_month
)
SELECT
    source_month,
    ride_id,
    rideable_type,
    started_at,
    ended_at,
    start_station_name,
    start_station_id,
    end_station_name,
    end_station_id,
    start_lat,
    start_lng,
    end_lat,
    end_lng,
    member_casual,

```

```

CASE
  WHEN ended_at > started_at      --creates the required ride_length--
  THEN ended_at - started_at      --if ride has valid end time after start time, calculate duration
  normally--
  ELSE null                      --but if end_at <= start_at, the new ride_length value
  become NULL--
  END AS ride_length,
  EXTRACT(DOW FROM started_at)::INTEGER + 1  --creates day of the week--
  AS day_of_week
FROM deduplicated;

```

B.9 — Verify the Cleaned Dataset

```

SELECT
  (SELECT COUNT(*) FROM divvy_trips_combined_raw) AS raw_rows,
  (SELECT COUNT(*) FROM divvy_trips_clean) AS clean_rows,
  (SELECT COUNT(*) FROM divvy_trips_combined_raw)
  - (SELECT COUNT(*) FROM divvy_trips_clean) AS rows_removed;

```

```

SELECT
  ride_id,
  COUNT(*) AS occurrences
FROM divvy_trips_clean
GROUP BY ride_id
HAVING COUNT(*) > 1;

```

```

SELECT
  COUNT(*) AS missing_ride_length  --there should be 29 now called NULL--
FROM divvy_trips_clean
WHERE ride_length IS NULL;

```

```

SELECT
  COUNT(*) AS negative_ride_lengths  --there are 0 negative durations--
FROM divvy_trips_clean
WHERE ride_length < INTERVAL '0 seconds';

```

Appendix C — Analyze Phase: SQL Analysis

C.1 — Confirm Analysis Population

```

SELECT
  COUNT(*) AS total_rides
FROM divvy_trips_clean;

```

```

SELECT
  member_casual,
  COUNT(*) AS total_rides,
  ROUND(
    100.0 * COUNT(*) / SUM(COUNT(*) OVER ()),
    2
  ) AS percentage_of_rides
FROM divvy_trips_clean
GROUP BY member_casual

```

```
ORDER BY total_rides DESC;
```

C.2 — Compare Ride Duration by Rider Type

```
SELECT
  member_casual,
  COUNT(ride_length) AS rides_with_valid_duration,
  ROUND(
    AVG(EXTRACT(EPOCH FROM ride_length) / 60.0),
    2
  ) AS avg_ride_minutes,
  ROUND(
    PERCENTILE_CONT(0.5)
    WITHIN GROUP (
      ORDER BY EXTRACT(EPOCH FROM ride_length) / 60.0
    )::numeric,
    2
  ) AS median_ride_minutes,
  ROUND(
    MIN(EXTRACT(EPOCH FROM ride_length) / 60.0),
    2
  ) AS min_ride_minutes,
  ROUND(
    MAX(EXTRACT(EPOCH FROM ride_length) / 60.0),
    2
  ) AS max_ride_minutes
FROM divvy_trips_clean
WHERE ride_length IS NOT NULL
GROUP BY member_casual;
```

C.3 — Compare Usage by Day of Week

```
SELECT
  member_casual,
  day_of_week,
  CASE day_of_week
    WHEN 1 THEN 'Sunday'
    WHEN 2 THEN 'Monday'
    WHEN 3 THEN 'Tuesday'
    WHEN 4 THEN 'Wednesday'
    WHEN 5 THEN 'Thursday'
    WHEN 6 THEN 'Friday'
    WHEN 7 THEN 'Saturday'
  END AS day_name,
  COUNT(*) AS total_rides,
  ROUND(
    100.0 * COUNT(*) /
    SUM(COUNT(*)) OVER (PARTITION BY member_casual),
    2
  ) AS percentage_of_total_rides
```

```

) AS pct_of_rider_type
FROM divvy_trips_clean
GROUP BY member_casual, day_of_week
ORDER BY member_casual, day_of_week;

```

C.4 — Compare Ride Duration by Day of Week

```

SELECT
  member_casual,
  day_of_week,
  CASE day_of_week
    WHEN 1 THEN 'Sunday'
    WHEN 2 THEN 'Monday'
    WHEN 3 THEN 'Tuesday'
    WHEN 4 THEN 'Wednesday'
    WHEN 5 THEN 'Thursday'
    WHEN 6 THEN 'Friday'
    WHEN 7 THEN 'Saturday'
  END AS day_name,
  ROUND(
    AVG(EXTRACT(EPOCH FROM ride_length) / 60.0),
    2
  ) AS avg_ride_minutes,
  ROUND(
    PERCENTILE_CONT(0.5)
    WITHIN GROUP (
      ORDER BY EXTRACT(EPOCH FROM ride_length) / 60.0
    )::numeric,
    2
  ) AS median_ride_minutes
FROM divvy_trips_clean
WHERE ride_length IS NOT NULL
GROUP BY member_casual, day_of_week
ORDER BY member_casual, day_of_week;

```

C.5 — Analyze Monthly Riding Patterns

```

SELECT
  DATE_TRUNC('month', started_at)::date AS month,
  TO_CHAR(DATE_TRUNC('month', started_at), 'Mon YYYY') AS month_name,
  member_casual,
  COUNT(*) AS total_rides
FROM divvy_trips_clean
WHERE started_at >= '2025-09-01'
  AND started_at < '2026-09-01'
GROUP BY
  DATE_TRUNC('month', started_at),
  member_casual
ORDER BY

```

```
month,  
member_casual;
```

```
SELECT  
  DATE_TRUNC('month', started_at)::date AS month,  
  TO_CHAR(DATE_TRUNC('month', started_at), 'Mon YYYY') AS month_name,  
  member_casual,  
  COUNT(*) AS total_rides,  
  ROUND(  
    100.0 * COUNT(*) /  
    SUM(COUNT(*)) OVER (PARTITION BY member_casual),  
    2  
  ) AS pct_of_rider_type  
FROM divvy_trips_clean  
WHERE started_at >= '2025-09-01'  
AND started_at < '2026-09-01'  
GROUP BY  
  DATE_TRUNC('month', started_at),  
  member_casual  
ORDER BY  
  month,  
  member_casual;
```

C.6 — Analyze Time-of-Day Usage

```
SELECT  
  EXTRACT(HOUR FROM started_at)::int AS hour_of_day,  
  member_casual,  
  COUNT(*) AS total_rides  
FROM divvy_trips_clean  
WHERE started_at >= '2025-09-01'  
AND started_at < '2026-09-01'  
GROUP BY  
  EXTRACT(HOUR FROM started_at),  
  member_casual  
ORDER BY  
  hour_of_day,  
  member_casual;
```

```
SELECT  
  member_casual,  
  CASE  
    WHEN EXTRACT(HOUR FROM started_at) BETWEEN 5 AND 10  
      THEN 'Morning'  
    WHEN EXTRACT(HOUR FROM started_at) BETWEEN 11 AND 15  
      THEN 'Midday'  
    WHEN EXTRACT(HOUR FROM started_at) BETWEEN 16 AND 19  
      THEN 'Evening'  
    ELSE 'Night'
```

```

    END AS time_of_day,
    COUNT(*) AS total_rides,
    ROUND(
        100.0 * COUNT(*) /
        SUM(COUNT(*)) OVER (PARTITION BY member_casual),
        2
    ) AS pct_of_rider_type
FROM divvy_trips_clean
WHERE started_at >= '2025-09-01'
AND started_at < '2026-09-01'
GROUP BY
    member_casual,
    time_of_day
ORDER BY
    member_casual,
    total_rides DESC;

```

C.7 — Compare Bike Type Usage

```

SELECT
    member_casual,
    rideable_type,
    COUNT(*) AS total_rides,
    ROUND(
        100.0 * COUNT(*) /
        SUM(COUNT(*)) OVER (PARTITION BY member_casual),
        2
    ) AS pct_of_rider_type
FROM divvy_trips_clean
WHERE started_at >= '2025-09-01'
AND started_at < '2026-09-01'
GROUP BY
    member_casual,
    rideable_type
ORDER BY
    member_casual,
    total_rides DESC;

```

C.8 — Compare Starting Station Patterns

```

WITH station_counts AS (
    SELECT
        member_casual,
        start_station_name,
        COUNT(*) AS total_rides
    FROM divvy_trips_clean
    WHERE start_station_name IS NOT NULL
        AND started_at >= '2025-09-01'
        AND started_at < '2026-09-01'

```

```

GROUP BY
    member_casual,
    start_station_name
),
ranked_stations AS (
    SELECT
        member_casual,
        start_station_name,
        total_rides,
        ROUND(
            100.0 * total_rides /
            SUM(total_rides) OVER (PARTITION BY member_casual),
            2
        ) AS pct_of_known_station_rides,
        ROW_NUMBER() OVER (
            PARTITION BY member_casual
            ORDER BY total_rides DESC
        ) AS station_rank
    FROM station_counts
)
SELECT
    member_casual,
    station_rank,
    start_station_name,
    total_rides,
    pct_of_known_station_rides
FROM ranked_stations
WHERE station_rank <= 10
ORDER BY
    member_casual,
    station_rank;

```

Appendix D — Supporting Tables

Table D.1 — Rider Type and Ride Duration Summary

member_casual	total_rides	percentage_of_rides
member	3,955,594	64.68%
casual	2,160,353	35.32%

member_casual	rides_with_valid_duration	avg Ride minutes	median Ride minutes	min Ride minutes	max Ride minutes
casual	2,160,344	20.7	10.86	0	1559.95
member	3,955,574	12.37	8.55	0	1559.9

Table D.2 — Rides by Day of Week

member_casual	day_of_week	day_name	total_rides	pct_of_rider_type
casual	1	Sunday	348,468	16.13%
casual	2	Monday	257,881	11.94%
casual	3	Tuesday	246,857	11.43%
casual	4	Wednesday	247,902	11.48%
casual	5	Thursday	268,025	12.41%
casual	6	Friday	334,557	15.49%
casual	7	Saturday	456,663	21.14%
member	1	Sunday	420,359	10.63%
member	2	Monday	566,150	14.31%
member	3	Tuesday	630,934	15.95%
member	4	Wednesday	631,953	15.98%
member	5	Thursday	625,494	15.81%
member	6	Friday	577,940	14.61%
member	7	Saturday	502,764	12.71%

Table D.3 — Ride Duration by Day of Week

member_casual	day_of_week	day_name	avg Ride Minutes	median Ride Minutes
casual	1	Sunday	24.29	12.38
casual	2	Monday	20.73	10.63
casual	3	Tuesday	18.04	9.73
casual	4	Wednesday	17.25	9.47
casual	5	Thursday	18.01	9.66
casual	6	Friday	20.21	10.67
casual	7	Saturday	23.18	12.64
member	1	Sunday	13.49	9.08
member	2	Monday	12.07	8.29
member	3	Tuesday	11.92	8.39
member	4	Wednesday	11.85	8.38
member	5	Thursday	11.91	8.37
member	6	Friday	12.31	8.41
member	7	Saturday	13.6	9.36

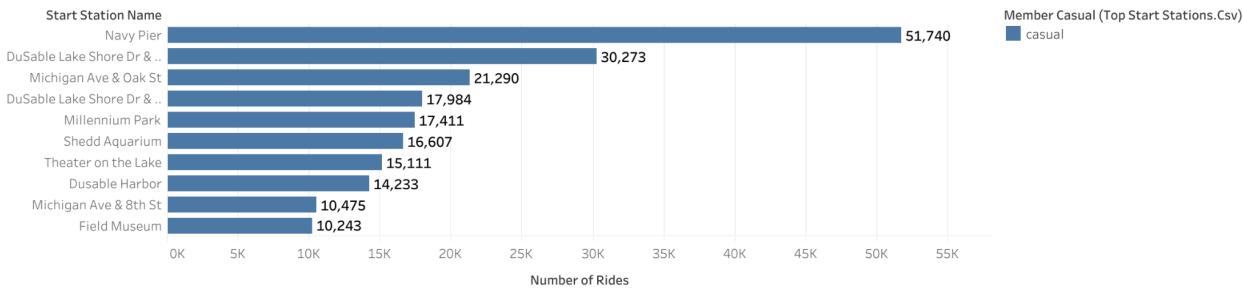
Table D.4 — Bike Type Usage

member_casual	rideable_type	total_rides	pct_of_rider_type
casual	electric_bike	1,589,951	73.60%
casual	classic_bike	570,174	26.40%
member	electric_bike	2,692,044	68.06%
member	classic_bike	1,263,510	31.94%

Table D.5 — Top Start Stations by Rider Type

member_casual	station_rank	start_station_name	total_rides	pct_of_known_station_rides
casual	1	Navy Pier	51,740	3.07%
casual	2	DuSable Lake Shore Dr & Monroe St	30,273	1.79%
casual	3	Michigan Ave & Oak St	21,290	1.26%
casual	4	DuSable Lake Shore Dr & North Blvd	17,984	1.07%
casual	5	Millennium Park	17,411	1.03%
casual	6	Shedd Aquarium	16,607	0.98%
casual	7	Theater on the Lake	15,111	0.90%
casual	8	Dusable Harbor	14,233	0.84%
casual	9	Michigan Ave & 8th St	10,475	0.62%
casual	10	Field Museum	10,243	0.61%
member	1	Canal St & Madison St	23,179	0.74%
member	2	State St & Chicago Ave	22,126	0.70%
member	3	Clinton St & Jackson Blvd	20,024	0.64%
member	4	Wells St & Elm St	19,638	0.62%
member	5	Clinton St & Madison St	19,477	0.62%
member	6	Wells St & Concord Ln	19,398	0.62%
member	7	Wells St & Huron St	17,923	0.57%
member	8	Wells St & Hubbard St	17,523	0.56%
member	9	Clark St & Elm St	16,650	0.53%
member	10	Navy Pier	16,289	0.52%

Top Starting Stations Differ Between Casual Riders and Members



Top Starting Stations Differ Between Casual Riders and Members

